

Jannet Trabelsi

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Portfolio: <https://jat0205.github.io/Portfolio/>
Codebase: <https://github.com/Jat0205/pittqlabsys-jannet-features>

EDUCATION

University of Pittsburgh, PA

| Expected April 2027

Bachelor of Science in Engineering Physics

Minors in Physics and Electrical Engineering

Certificates in Photonics | Quantum Computing and Quantum Information

Bachelor of Philosophy (honors thesis): Adiabatic DEER Quantum Sensing with Single NV Centers

GPA 3.912/4.000

Charles University, Prague, Czech Republic

| Summer 2025

Engineering study abroad program

RESEARCH INTERESTS

Experimental quantum information and coherent qubit control — with hands-on expertise in laser and microwave control, pulse sequencing, IQ calibration, and sub-nanosecond timing developed through NV-center work — aiming toward experimental trapped-ion quantum computing and scalable quantum control.

MANUSCRIPT

First-author manuscript on Adiabatic NV DEER experiment, in preparation

| Expected Fall 2026

RESEARCH EXPERIENCE

Dutt Lab, University of Pittsburgh

| January 2024 – Present

Pulsed Optically Detected Magnetic Resonance Experiment

- Developed a Python-based pulse sequencing and calibration system achieving 0.8 ns timing resolution for multi-channel laser and microwave control, enabling high-fidelity NV-center ODMR measurements
- Architected a State Machine ADbasic Program for T1, Rabi, Ramsey, and Hahn echo measurements
- Prepared and wire-bonded nanodiamond and bulk diamond samples for ODMR
- Made IQ calibration to do pulsed ODMR by programming SG384, spectrum analyzer, and Proteus AWG
- Programmed a MultiHarp-based time-correlated photon-counting experiment for gate alignment, hardware jitter/drift detection, and offline gating and analysis to increase contrast.

Continuous Wave Optically Detected Magnetic Resonance Experiment

- Debugged and validated the CW ODMR experiment, identifying and fixing errors that were preventing operation
- Achieved CW ODMR with Single NV centers in diamond with ZFS and magnetic field splitting

Confocal Microscopy and Positioning Experiment

- Built a Python suite to automate and optimize NV-center count collection in confocal microscopy
- Designed a modular Python GUI for sample positioning and data analysis, enabling reproducible confocal scans
- Developed an automated EMCCD camera scan with microdrive and implemented wide-field optics and white-light imaging that acquires the full field in a single exposure ~ 360 times faster than a point-by-point scan
- Performed confocal alignment and optimization for NV-center imaging; cleaved samples and prepared nanodiamond solutions; collected photoluminescence data via spectrometer

Lab Software Project

- Developed a Python-based instrument control suite (FastAPI servers, spectrum analyzer GUI, and micro-stage automation), enabling unified data collection, data saving, and repurposing lab hardware

Spin Charge Conversion Experiment

- Aligned and tuned a dye ring laser cavity
- Designed and aligned optical setups for spin charge conversion experiment

Annealing Chamber Project

- Designed and prototyped an annealing chamber, performed computational fluid dynamics and thermal simulations using SOLIDWORKS, and communicated instructions with machinists

Levitating Diamond Experiment

- Designed a vacuum re-entrant flange and performed static simulations for UHV window using SOLIDWORKS to use a higher Numerical Aperture microscope objective, improving imaging of levitated diamond by 43%
- Developed an Arduino-based PID control system for laser thermal stabilization in a levitating diamond setup

Lab Training

- Authored SOPs and training guides and trained graduate students, improving onboarding and lab safety

HONORS

TRIO 2024 National Scholarship Award

| October 2024

Gloria and Mark Tressler Scholarship for the Swanson School of Engineering

| October 2025 – April 2027

Swanson School of Engineering Honor Student

| August 2024 – Present

Dean's List

| August 2023 – Present

Sigma Pi Sigma Physics Honor Society

| Spring 2026 – Present

TECHNICAL EXPERTISE

Instrumentation & Control

- Python (Instrument Control, Automation, GUIs), ADbasic, FastAPI Servers, GPIB/USB/LAN/RS232/PXI Instrument Control, PID Loop, Waveform Generation

Hardware & Electronics

- Altium PCB Design, LTspice, Digital/Analog Circuit Design, VHDL, Quartus, Soldering

Simulation & CAD

- SOLIDWORKS (CFD, Static, Thermal), ANSYS Mechanical, Granta EduPack, Autodesk

Quantum & Optics Experiments

- Confocal Microscopy, NV Centers, AOMs, Wavemeters, Spectrometers, Cryostats, Vacuum Systems, Interferometry, Annealing Chamber, Arduino, Actuators, TEC, Universal Counters, Spectrum Analyzer, EMCCD Camera, Arbitrary Waveform Generators

Programming

- Python, C++, MATLAB, ADbasic, JSON, HTML, Git

CLASS PROJECTS

Quantum Directed Research, University of Pittsburgh

| January 2026 – Present

- Applied group theory to analyze defect symmetries in SiC and NV0/NV– centers (directed study with Dr. Devaty)

Optics, University of Pittsburgh

| November 2025

- Executed optical pumping experiment with a Nd:YAG crystal, including alignment, data acquisition, and analysis
- Prepared Michelson interferometer and measured coherence length and time for various light sources.
- Investigated single-photon interference phenomena through measurement using a PMT and quantitative analysis

ECE Junior Design

| January 2026 – April 2026

- Designed, prototyped, built, and tested three PCB projects: educational dummy spectrometer, Quantum Bop It to teach Quantum Computing in a team environment, and hybrid analog-digital AOM RF Driver to modulate laser

LEADERSHIP AND COMMUNITY ENGAGEMENT

Volunteer, Quantum Computing Outreach Event

| September 2024 – September 2025

- Organized quantum outreach events, engaging 19 students through interactive sessions
- Led activities including quantum computing games and dye laser demo to inspire interest in the quantum field

LANGUAGES

Arabic: Native

English: Professional / Fluent

French: Professional / Fluent